

Miny

Na čem celá hra stojí?

field

```
[[0 0 1 1 2 2 3 2 1 0 0 1 9 9 9 1]
[0 0 1 9 2 9 9 9 2 0 0 1 3 4 3 1]
[0 0 1 1 2 2 4 9 3 2 2 2 2 9 2 1]
[0 0 0 1 2 2 2 2 3 9 9 3 9 3 9 2]
[1 1 1 1 9 9 2 2 9 4 9 4 2 3 2 9]
[1 9 1 1 3 3 3 9 2 2 1 2 9 1 1 1]
[1 1 1 0 1 9 2 1 1 0 1 2 2 1 1 1]
[0 0 0 0 1 1 1 0 0 0 1 9 1 1 2 9]
[0 0 0 0 0 0 0 0 0 0 2 2 2 2 9 3]
[0 0 0 0 0 1 1 1 0 0 1 9 2 3 9 2]
[0 0 1 1 1 1 9 3 2 1 1 2 3 9 3 2]
[1 1 1 9 1 2 3 9 9 1 0 1 9 4 9 1]
[9 1 1 1 1 1 9 3 2 1 0 1 2 9 2 1]]
```

+

bool field

```
[[1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0]
[1 1 1 2 1 0 0 0 0 0 0 0 0 0 0 0]
[1 1 1 1 1 1 1 2 1 0 0 0 0 0 0 0]
[1 1 1 1 1 1 1 1 1 2 2 0 0 0 0 0]
[1 1 1 1 2 2 1 1 2 1 2 0 0 0 0 0]
[0 2 1 1 1 1 1 2 1 1 1 0 0 0 0 0]
[1 1 1 1 1 2 1 1 1 1 1 0 0 0 0 0]
[1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0]
[1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0]
[1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0]
[1 1 1 1 1 0 1 1 1 1 1 0 0 0 0 0]
[1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0]
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0]]
```

=

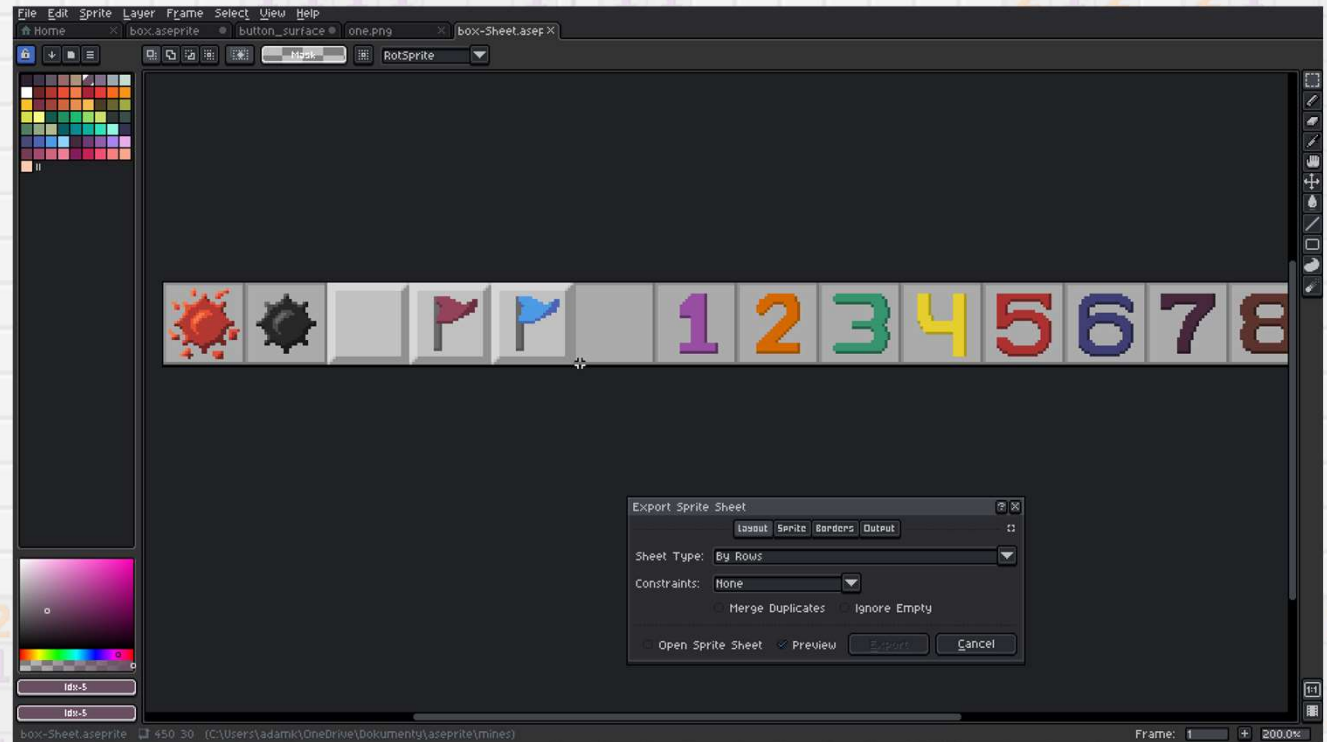


Jak to přenést do podoby hry?



Grafická stránka – proč zrovna pixelart?

- známá velikost, typický “minesweeper” styl
- snadná vlastní tvorba
- Aseprite



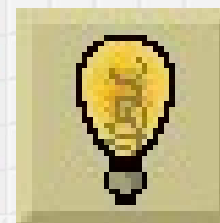
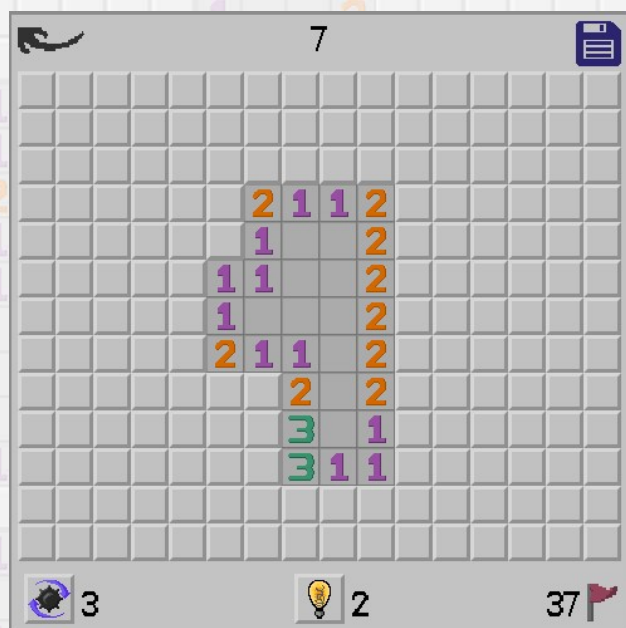
Herní mechanika

- left-click, right-click, middle-click



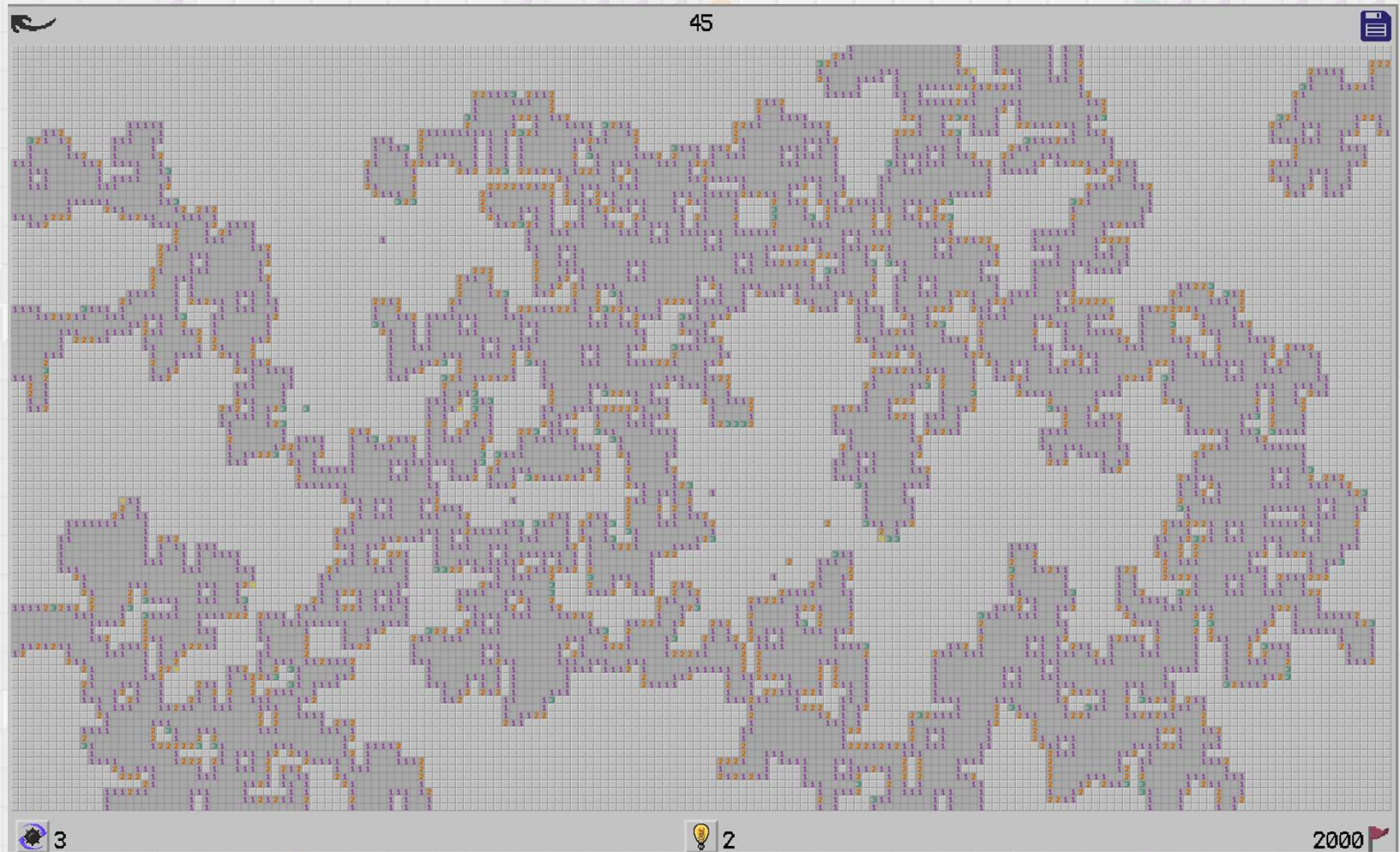
Vliv náhodné generace na průběh hry a řešení

- first click, hint

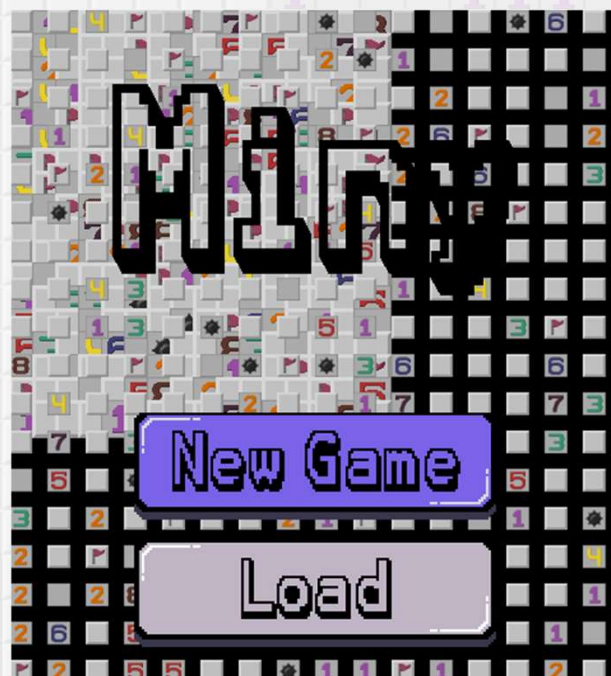


-
- | | | | | | | | | | |
|---|---|--|--|--|--|--|--|--|--|
| 2 | ♥ | | | | | | | | |
| 3 | ♥ | | | | | | | | |
| 3 | ♥ | | | | | | | | |
| 3 | ♥ | | | | | | | | |
| 3 | ♥ | | | | | | | | |
| 3 | ♥ | | | | | | | | |
| 3 | ♥ | | | | | | | | |
| 3 | ♥ | | | | | | | | |
| 3 | ♥ | | | | | | | | |
| 2 | ♥ | | | | | | | | |

Sandbox a scale factor



a co všechno to může rozbít



Ukládání pole

```
1 {  
2   "boolField": [[0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0],  
3   [0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0],  
4   [1, 1, 1, 1, 1, 1, 1, 1, 2, 0, 1],  
5   [1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1],  
6   [1, 1, 1, 1, 0, 2, 1, 1, 1, 0, 1],  
7   [1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 1],  
8   [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1],  
9   [1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1],  
10  [1, 1, 2, 1, 1, 1, 2, 1, 1, 1, 1],  
11  [1, 1, 0, 3, 0, 0, 1, 2, 1, 1, 1]],  
12  "field": [[2, 3, 2, 1, 0, 0, 0, 0, 0, 2, 9],  
13  [9, 9, 9, 1, 0, 0, 0, 1, 2, 2, 3],  
14  [2, 3, 2, 1, 0, 0, 0, 1, 9, 9, 3],  
15  [0, 0, 0, 1, 2, 2, 1, 1, 3, 9, 2],  
16  [0, 0, 0, 1, 9, 9, 1, 0, 2, 2, 2],  
17  [0, 0, 0, 1, 2, 2, 1, 0, 1, 9, 1],  
18  [0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1],  
19  [0, 1, 1, 1, 0, 1, 1, 1, 0, 0, 0],  
20  [0, 2, 9, 4, 3, 3, 9, 2, 1, 0, 0],  
21  [0, 2, 9, 9, 9, 9, 3, 9, 1, 0, 0]],  
22  "timePlayed": 10,  
23  "reshuffleCount": 2,  
24  "hintCount": 1,  
25  "mineCount": 18,  
26  "difficulty": "Easy",  
27  "remainingFlags": 12,  
28  "scaleFactor": 1.0  
29 }
```



Databáze výsledků

```
182 }
183 },
184 "(30, 30, 200)": {
185   "gold": {
186     "username": "gold",
187     "time": "327"
188   },
189   "silver": {
190     "username": "silver",
191     "time": "589"
192   },
193   "bronze": {
194     "username": "bronze",
195     "time": "982"
196   }
197 },
198 "(10, 10, 30)": {
199   "gold": {
200     "username": "gold",
201     "time": "50"
202   },
203   "silver": {
204     "username": "silver",
205     "time": "91"
206   },
207   "bronze": {
208     "username": "bronze",
209     "time": "152"
210   },
211   "rGDHAedJYz2EYQQIm0FS": {
212     "username": "Sir Meowsington IV",
213     "time": "190"
214   }
215 }
```

Leaderboards	
1. gold	50s
2. silver	91s
3. bronze	152s
4. Sir Meowsington IV	190s